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**RADIOMICS ANALYSIS AFTER OBTAINING THE REGION OF
INTEREST MASK:
A GUIDE FOR RADIOLOGISTS**

Methodological Recommendations No. 80

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These methodological recommendations are intended for use in daily practice by radiologists for the extraction of radiomics features.

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TERMS AND DEFINITIONS

The following terms with their respective definitions are applied in this document:

Voxel – The smallest logical element of a three-dimensional digital image in raster graphics. Voxels are analogues of two-dimensional pixels for three-dimensional space.

Region of Interest (ROI) – A defined area of an image or dataset that is the object of analysis.

Pixel (short for *picture element*) – The smallest logical element of a two-dimensional digital image in raster graphics. A pixel is an indivisible object of rectangular or circular shape, characterized by a certain color. A raster computer image consists of pixels arranged in rows and columns.

Radiomics – A method that extracts a large number of features from medical images using specialized data characterization algorithms. The method lies at the intersection of radiology, computer science, and mathematical statistics. The term "texture analysis" can be used as a synonym.

Radiomics Feature – A quantitative characteristic of an image obtained as a result of radiomics analysis.

Gray Levels (shades of gray, gray gradations) – An image color mode implemented in the form of a table (Grayscale) indicating the white brightness for each point.

SYMBOLS AND ABBREVIATIONS

- **PC** – Principal Component (*ГК – главная компонента*)
- **GLCM** – Gray Level Co-occurrence Matrix
- **GLDM** – Gray Level Dependence Matrix
- **GLRLM** – Gray Level Run Length Matrix
- **GLSZM** – Gray Level Size Zone Matrix
- **NGTDM** – Neighbouring Gray Tone Difference Matrix
- **ROI** – Region of Interest
- **RMSE** – Root-Mean-Square Error

INTRODUCTION

Radiomics is a promising method of modern medicine aimed at improving the quality of diagnosis and therapy planning using advanced medical image analysis techniques. Radiomics analysis is conducted based on the extraction and processing of medical image parameters—radiomics features that are invisible to the human eye.

Radiomics allows for the quantitative analysis of textural information obtained via the mathematical evaluation of the spatial distribution of signal intensities and pixel interrelationships. Furthermore, radiomics can quantitatively evaluate visually observable differences in image intensity, shape, or texture, reducing the subjective nature of medical image interpretation across Computed Tomography (CT), Magnetic Resonance Imaging (MRI), Ultrasound (US), and Positron Emission Tomography (PET).

The long-term outlook of the method involves using it as a supportive tool for forecasting and decision-making regarding treatment strategies. Radiomics has found its widest application in oncology. It does not imply the automation of diagnostic processes, but rather provides additional data. Radiomics analysis can be useful for various clinical tasks: diagnosis, staging, and assessment of treatment response.

In recent years, research has been actively conducted on the relationship between radiomics features and gene expression (**radiogenomics**) as well as immunohistochemical features, which will help improve treatment response predictions in the future. **Delta-radiomics** introduces a temporal component, involving the extraction of quantitative features from sets of images acquired during a course of treatment to provide information on feature evolution.

Radiomics analysis has several limitations. The main one is the lack of standardized methods and, consequently, low reproducibility of results. Virtually every study selects its own radiomics features and builds a custom model. The non-reproducibility of significant radiomics features between studies is due to the lack of standardization in image acquisition/processing and varying methods of feature extraction/selection.

Currently, attempts are being made to overcome this limitation by creating a standardized database of radiomics features, such as the **IBSI (Image Biomarker Standardisation Initiative)** project. The main stages of texture analysis include: medical image acquisition, segmentation of the region of interest on the acquired images, radiomics feature extraction, creation of a prognostic model based on the extracted and selected features, model validation on internal and external data, evaluation of the model's predictive power, and its further refinement using specialized tools.

The specialized skills of radiologists are required during the stages of image acquisition, labeling, and radiomics feature extraction. The stages of creating and testing the prognostic model require the specialized skills of medical statisticians. Nevertheless, we recommend that radiologists—the end-users of the developed models—participate in all stages of prognostic model creation to facilitate the interpretation of the results.

The purpose of these recommendations is to describe the main stages and present the key concepts of radiomics analysis. To achieve this, a detailed characterization of radiomics features is given, a step-by-step procedure for their extraction is described, and an example of statistical analysis of the obtained data is provided (Appendix A). Despite the abundance of literature in the field of radiomics, this work is the first methodological guide on radiomics analysis for a Russian-speaking audience.

EXTRACTION AND ANALYSIS OF RADIOMICS FEATURES

Image Labeling Software

Before starting radiomics analysis, it is necessary to label the images by isolating the regions of interest. Table 1 provides summary information on the general characteristics of modern software tools for annotating digital diagnostic images.

Table 1 – General characteristics and file handling in software for digital diagnostic image annotation

Function/ Characteristic	3D Slicer	ITK SNAP	MITK	Medseg	Supervisely	CVAT
Open source	Yes	Yes	Yes	No	No	Yes
Web application	No	No	No	Yes	Yes	Yes
Local installation	Yes	Yes	Yes	No	In extended	Yes
Custom modules	Yes	Yes	Yes	No	Yes	Yes
Installation difficulty	Low	Low	Low	—	—	High
Min RAM / CPU	4/8 GB, 4/8 cores	2/4 GB	—	—	8/32 GB	—
OS Support	Win, Mac, Linux	Win, Mac, Linux	Win, Mac, Linux	—	Linux	Win, Mac, Linux
GPU Recommendation	Recommended	Recommended	—	Chrome, GPU	200GB SSD, GPU	Recommended
Medical format support	+	+	+	—	—	—
Batch load image/seg	+	+	+	—	+/-	—
Multi-segmentation/ image	+	+/-	+	—	—	—

Note: "—" — inapplicable/info missing; "+" — present; "-" — absent; "+/-" — partially implemented.

Thus, among the freely available software, **3D Slicer** possesses the most comprehensive functionality for image labeling.

Radiomics Feature Extraction Software

A comparative analysis of popular free software tools for extracting radiomics features was performed: PyRadiomics, LIFEx, MaZda, and IBEX (Table 2).

Table 2 – General characteristics of radiomics feature extraction software

Characteristic	PyRadiomics	LIFEx	MaZda	IBEX
Free Program	Yes	Yes	Yes	Yes
Open Access	Yes	No (Email reg)	No (License req)	No (Dropbox account)
Last Update	2023 (1-2 yrs)	2024 (1-2/yr)	2009 (End)	2017 (End)
Citations (Google Scholar)	4157	794	743	373
Total Features Count	107	44	352 (2D) / 892 (3D)	91
Math Equation Doc	Yes	Yes	No	No
IBSI Compliance	Yes	Yes	No	No
Supported Formats	NIFTI-1, JPG, TIFF, PNG	DICOM, NIFTI-1, JPG, etc.	DICOM	DICOM, BMP
Language / Integration	Python / Slicer GUI	Java / No	C++ / No	Matlab / No

PyRadiomics is the most popular program for this task. Unlike other free programs, it is completely open-access and does not require registration. A major advantage of PyRadiomics is its calculation of radiomics features compliant with the IBSI initiative, which is vital for standardization, along with its support for a wide range of medical image formats.

General Characteristics of Radiomics Features

Traditionally, three main groups of radiomics features are identified: first-order features, second-order features, and shape features.

- **First-order features (Histogram analysis):** Extract pixel intensity values within the ROI, which are then displayed graphically as a histogram. Based on the histogram, metrics including uniformity, variance, asymmetry, and randomness of pixel intensity values within the ROI are computed. The most common metrics are: mean, standard deviation, skewness, kurtosis, entropy, and energy.
- **Second-order features:** Examine the spatial relationships between pixels within the ROI. By evaluating them, one can obtain additional information regarding pixel signal intensity variability across smoother, uniform regions (lower texture variability) or heterogeneous regions (higher texture variability).
- **Shape features:** Depend not on gray intensity, but on the geometric characteristics of the ROI (e.g., surface area, mesh volume, compactness).

Use of Post-processing Filters

Radiomics features can be extracted from both original and filtered images. PyRadiomics enables the use of wavelet-based features. These post-processing filters are applied to increase the number of extracted texture metrics and remove noise. The wavelet transform uses a set of special filters to recursively decompose the original image into low-frequency and high-frequency sub-bands along the x, y, and z axes, yielding eight new images of the same size from a single original image.

Installation and Extraction Pipeline

The sequence for extracting radiomics features in SlicerRadiomics for a single study is as follows:

1. Open the **Informatics** tab and select the **Radiomics** extension.
2. In the IO tab, select the analyzed series (**Input Image Volume**, e.g., PET or CT) and select the image mask (**Input regions**).
3. In the Features tab, select the feature classes to be extracted (**firstorder**, **glcm**, **gldm**, **glszm**, **ngtdm**, **shape**, **shape 2D**).
4. If necessary, check the box for **Wavelet-based features**.
5. Set the **binWidth** parameter (determines the histogram bin width for first-order metrics). The default value is 25. It is critical to maintain a consistent binWidth across all patients within a study to ensure identical calculation settings.
6. Click **Apply** to run the computations. Export the generated table as a **.CSV** file, parse it inside Microsoft Excel using the *Text to Columns* delimiter feature, and compile patient profiles.

CONCLUSION

Radiomics is a highly promising medical diagnostics method capable of becoming a standard routine tool for clinicians. User-friendly, open-access instruments are available for radiologists to extract radiomics features independently. These recommendations serve as a comprehensive, step-by-step instruction manual for extracting

radiomics features from medical images using 3D Slicer and its SlicerRadiomics extension. Based on the comparative analysis performed, **PyRadiomics** stands out as the most widely used and preferred tool at present.